

REVIEW

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Advanced strategy for cancer detection based on volatile organic compounds in breath

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Abstract

The analysis of volatile organic compounds (VOCs) in exhaled breath has emerged as a promising non-invasive approach for cancer diagnosis, offering advantages in speed, safety, cost-effectiveness, and real-time monitoring. Two primary methodologies are employed for VOCs detection: mass spectrometry (MS)-based techniques, which provide high-precision identification and quantification of individual compounds, and sensor-based pattern recognition methods, which detect disease-specific VOC signatures. Despite their diagnostic potential, inconsistencies in accuracy highlight the need for a comprehensive evaluation of these techniques. This review synthesizes evidence from clinical studies through meta-analysis to assess the diagnostic performance of MS and sensor-based approaches. Furthermore, we examine variations in VOC profiles across cancer types, which may influence diagnostic precision, and discuss key biomarkers, analytical methodologies, current challenges, and future directions in VOCs-based diagnostics. Meta-analysis revealed a high diagnostic accuracy, with a mean area under the receiver operating characteristic curve (AUC) of 0.94 (95% CI 0.91–0.96), sensitivity of 89% (95% CI 87%–90%), and specificity of 87% (95% CI 84%–88%). Notably, no significant difference was observed between MS and sensor-based methods (AUC: 0.91 vs. 0.93, $p=0.286$), supporting the potential of sensor technologies for clinical application. Subgroup analysis further indicated no statistical difference in AUCs between heterogeneous and homogeneous sensor groups, suggesting that simplified detection systems may be feasible. Despite these promising results, standardization of protocols and methodological consistency remain critical challenges. Future efforts should focus on large-scale, well-designed clinical trials to validate and optimize VOCs-based breath tests, enhancing their diagnostic reliability and translational potential in oncology.

Keywords Cancer diagnostics, Volatile organic compound, Breath analysis, Non-invasive biomarker, Mass spectrometry, Sensor

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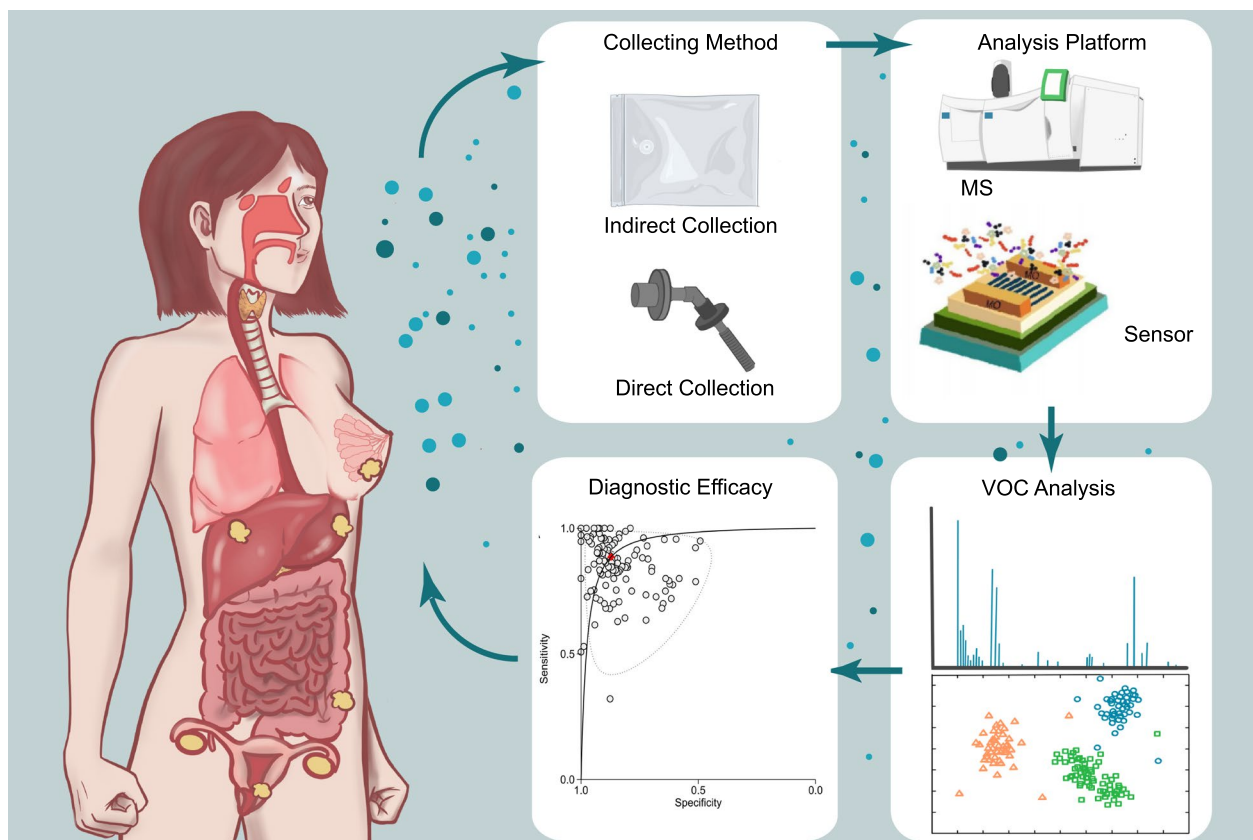
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Graphical Abstract



Introduction

Annually, over 20 million individuals are diagnosed with cancer, resulting in 10 million deaths, representing about one-sixth of total global deaths [1]. Improving early detection methods is imperative to facilitate timely intervention and improve outcomes [2]. Prevailing diagnostic modalities are frequently characterized by their invasiveness, reliance on radiative techniques, susceptibility to operator variability, and a dependence on the discernible stage and size of neoplasms [3]. These constraints underscore an exigent demand for alternative diagnostic measures that are both non-invasive and objective [4, 5]. Amongst such emerging technologies, volatile organic compounds (VOCs)-based breath testing stands at the forefront, offering a paradigm shift towards precise and non-invasive diagnostic capabilities.

Exhaled breath VOCs hold potential for early cancer detection through metabolic fingerprinting and advanced analytical methodologies. VOCs in exhaled breath are reflective of metabolic processes that may be altered in the presence of cancer cells [6, 7], providing a "metabolic

fingerprint" that can be detected with high precision [8, 9]. This biochemical insight underscores the rationale for summarizing current VOCs-based technologies, as it is the intricacies of these metabolic changes, coupled with advanced detection techniques, that facilitate the early identification of malignancies [10–13]. Two primary analytical methodologies have been developed for the evaluation of VOCs present in exhaled breath [14]. The first method involves an analytical chemistry approach utilizing mass spectrometry (MS), with or without spectrometric techniques, to identify and quantify the chemical composition of VOCs in a given sample [15]. This method allows for the individual examination of each compound and the assessment of the likelihood that any quantifiable compound could serve as a biomarker for a particular disease. The second method employs semi-selective/cross-reactive sensors in conjunction with pattern recognition techniques, aiming to detect patterns of VOCs that act as unique "fingerprints" for a specific disease [16]. Although this methodology may not facilitate the identification of each individual VOCs, it has

the potential to uncover overarching patterns that could assist in recognizing analogous cases or in developing specific designs informed by the reported VOCs [17].

VOCs-based breath tests have demonstrated considerable potential in clinical oncology, as evidenced by multiple studies [18–20]. However, significant heterogeneity in reported diagnostic accuracy underscores the need for a critical reappraisal of existing evidence [21]. A systematic meta-analysis is therefore warranted to consolidate findings from recent clinical trials and provide a robust evaluation of test efficacy. Furthermore, while MS remains the gold standard for cancer detection, comparative data assessing its diagnostic performance relative to sensor-based pattern recognition approaches remain scarce. This gap necessitates subgroup analyses to delineate differences in diagnostic accuracy between methodologies. Additionally, the inherent variability in VOC signatures across different cancer types contributes to observed disparities in test performance, highlighting the need for disease-specific biomarker validation [22]. Summarizing the landscape of exhaled VOCs and evaluating VOCs-based tests' accuracy for different cancers is crucial for refining the application and effectiveness of breath tests.

Herein, we aim to: (1) summarize the landscape of exhaled VOCs across a variety of cancers; (2) conduct a comprehensive review of the technology and methodology used throughout the breath testing process in cancer diagnosis; (3) assess and compare diagnostic accuracies within different breath detection methods; and (4) propose a framework and paradigm for the design of future trials aimed at achieving higher quality outcomes.

Study selection

Study selection

The literature search yielded 34,739 records (Supplementary Fig S1). Duplicates and studies with unrelated titles, abstracts, or full texts were excluded. Among 245 studies with full texts, 65 studies were precluded based on the exclusion criteria. Consequently, 180 researches were included for qualitative synthesis. Additionally, for the meta-analysis, 126 studies were included. Two studies utilized the same cohort [23, 24]; the one employing Gas chromatography-mass spectrometry (GC-MS) was selected for the meta-analysis [23]. The Deeks test for funnel plot symmetry revealed the absence of any significant publication bias (bias = 5.41, $p = 0.182$). Supplementary Fig S2 displayed the corresponding Deeks plot, also known as funnel plot.

Study characteristics

The systematic review encompassed MS ($n = 128$) and sensors ($n = 52$), with the cancer patient sample size ranging from 10 to 1,051 (Supplementary Table S1).

The included studies examined various types of cancer, including lung cancer ($n = 100$), breast cancer ($n = 24$), gastroesophageal cancer ($n = 22$), colorectal cancer ($n = 11$), head and neck cancer [$n = 11$ (head and neck squamous cell carcinoma = 9, thyroid cancer = 2)], liver cancer ($n = 7$), malignant pleural mesothelioma ($n = 5$), oral cancer ($n = 4$), ovarian cancer ($n = 3$), prostate cancer ($n = 2$), pancreatic cancer ($n = 2$), cervical cancer ($n = 1$), cholangiocarcinoma ($n = 1$), chondrosarcoma ($n = 1$), and malignant mesothelioma ($n = 1$). Several studies included patients with more than one cancer type [25–27].

The meta-analysis incorporated data from 5578 patients and 9402 healthy controls for MS detection. 2551 patients and 3668 healthy controls were included in the meta-analysis of sensor detection. Certain studies employed both MS and sensors and distinguished between cancer patients and healthy individuals, subsequently analyzing the results as two separate articles. For the two studies that compared various cancer types with a single cohort of healthy controls, the low heterogeneity permitted individual meta-analytic evaluations for each cancer type separately [26, 28]. Tests differentiated cancer patients from those with other non-neoplastic diseases of the same organs, e.g., between lung cancer and COPD patients, were excluded from this analysis due to heterogeneity between studies.

VOCs sources and markers for cancer diagnosis

The hypothesis that VOCs can be used in cancer diagnosis

Cancer-related metabolic alterations influence VOC profiles, offering potential biomarkers detectable in breath. Metabolic alterations are continuously occurring within the body, both during normal physiological pathological processes. Tumor proliferation is typically associated with alterations in gene expression and/or protein composition [29]. Hypoxia, the hyperproliferation of cells, heightened inflammatory responses, and increased activity of reactive oxygen species, along with other cancer-associated pathological mechanisms, lead to significant alterations in the spectra and concentrations of VOCs, both locally and systemically, which is subsequently released into bodily fluids [30, 31]. In the realm of basic science, numerous spectroscopic and spectrometric techniques have been employed to conduct *in vivo*, *in vitro*, and *ex vivo* experiments aimed at elucidating the characteristics of VOC biomarkers. These methodologies have yielded hundreds of preliminary identifications of VOCs released directly by cultured cells and tissues, independent of systemic changes that may arise from secondary mechanisms and hemodynamic factors [32]. While the biochemical mechanisms related to specific VOCs are predominantly theoretical at this stage, numerous studies have indicated that the VOCs present in the exhaled

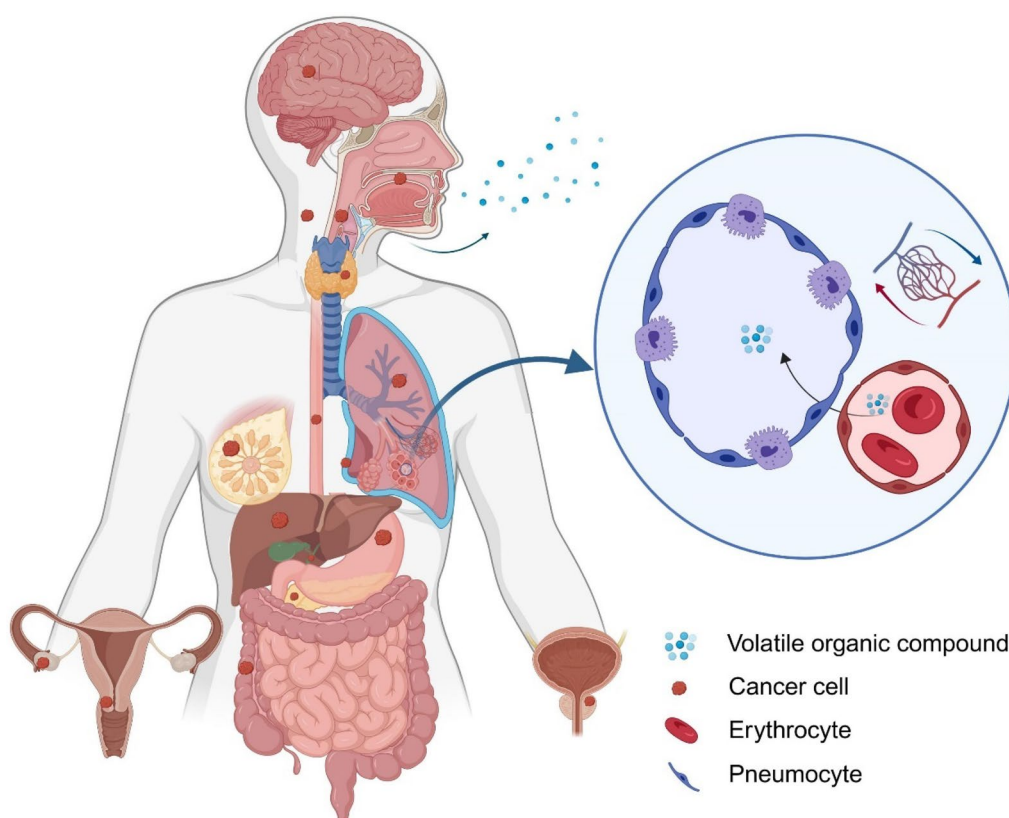


Fig. 1 The principle of VOCs detection and the application of VOCs detection in different tumor types

breath of cancer patients exhibit significant differences when compared to those found in healthy individuals [33, 34]. Based on the aforementioned information, it is a plausible hypothesis that the atypical VOCs associated with cancer are generated by cancer cells, and these compounds are subsequently released into the endobronchial cavity, from which they are transferred and eliminated through various bodily fluids (Fig. 1).

In a standard sample of exhaled breath, it is typical to detect approximately 3000 distinct VOCs which are exogenous or endogenous in varying concentrations, typically ranging from parts per trillion by volume (pptv) to parts per billion by volume (ppbv) [35]. The exogenous VOCs can enter the body through inhalation or skin absorption, originating from solvents and petroleum-based products. In contrast, endogenous VOCs are generated internally through cellular biochemical processes. There is notable diversity in the makeup of VOCs across individuals, with a limited subset of VOCs being linked to a prevalent health condition within a particular demographic [30].

Reported VOC markers for cancer diagnosis

Various types of VOCs can serve as biomarkers in the exhaled breath of cancer patients (Supplementary

Table S2 and S3) [19, 20]. These categories include alkanes, alcohols, aldehydes, ketones, nitriles and aromatic compounds [32]. While the exact origin remains uncertain, various potential biochemical routes have been proposed. One such example is the generation of alkanes as a result of oxidative stress within the cancer microenvironment, triggered by hypoxic or inflammatory conditions [31]. Unsaturated hydrocarbons, such as isoprene, are also present in exhaled breath and are produced through the mevalonic pathway of cholesterol synthesis. Over-activation of cytochrome P450 enzymes in cancer patients might elevate alcohol levels. In addition to these enzymes, cancer is distinguished by elevated levels of alcohol dehydrogenase and aldehyde dehydrogenase, which play crucial roles in the metabolism of aldehydes [30, 32]. An overabundance of cell growth due to reduced oxygen levels triggers anaerobic respiration, during which the glycolytic pathway generates energy and produces ketones and alcohols [31]. Sulfur-containing compounds in exhaled breath may arise from the incomplete metabolism of methionine in the transamination pathway. VOCs permeate through the cancer cell membrane and enter the bloodstream, eventually being exhaled. The precise origins of numerous endogenous

Table 1 VOCs that have been consistently identified by various research groups within a specific type of cancer

Cancer type	Compound name	Molecular weight	Chemical class	Reference
Gastric	2-methylbuta-1,3-diene	68.12	Hydrocarbon, aliphatic	[36, 37]
	Propan-2-one	86.13	Ketone	[36, 38]
	Hexadecane	226.44	Hydrocarbon, aliphatic	[39, 40]
	1,3-dioxolan-2-one	88.06	Ketone	[38, 39]
	1,2,3-trimethylbenzene	120.19	Hydrocarbon, aromatic	[38, 40]
	Butan-2-one	72.11	Ketone	[40, 41]
Colorectal	Tetradecane	198.39	Hydrocarbon, aliphatic	[42, 43]
	Ethylbenzene	106.16	Hydrocarbon, aromatic	[42, 43]
	Toluene	92.14	Hydrocarbon, aromatic	[42, 43]
	Benzaldehyde	106.12	Aldehyde	[42, 43]
	Dodecane	170.33	Hydrocarbon, aliphatic	[42, 44]
	Acetic acid	60.05	Carboxylic acid	[26, 42]
	Cyclohexanone	98.14	Ketone	[43, 44]
	4-ethyloct-1-yn-3-ol	154.25	Alcohols/phenols	[44, 45]
	1,3-xylene	106.16	Hydrocarbon, aromatic	[25, 46]
	1,4-xylene	106.16	Hydrocarbon, aromatic	[43, 46]
Breast	Ethanol	46.07	Alcohols/phenols	[47, 48]
	Heptanal	114.19	Aldehyde	[49, 50]
	Cyclohexanone	98.14	Ketone	[51, 52]
Liver	2-ethylhexan-1-ol	130.229	Alcohols/phenols	[52, 53]
	Acetonitrile	41.05	Nitrogen compound	[54, 55]
MPM	Benzonitrile	103.12	Nitrogen compound	[23, 56]
Lung	Propan-1-ol	60.1	Alcohols/phenols	[57–62]
	Butanal	72.11	Aldehyde	[58, 63–67]
	Pentane	72.15	Hydrocarbon, aliphatic	[26, 68–72]
	Pentanal	86.13	Aldehyde	[64, 66, 67, 73–75]
	2-methylbuta-1,3-diene	68.12	Hydrocarbon, aliphatic	[58, 61, 66, 74–76]
	Cyclohexanone	98.14	Ketone	[61, 62, 67, 72, 77–79]
	Toluene	92.14	Hydrocarbon, aromatic	[25, 58, 61, 66, 71, 75, 80]
	Propan-2-one	58.08	Ketone	[57, 58, 61, 62, 67, 76, 81]
	Butan-2-one	72.11	Ketone	[57, 61, 65, 67, 81–83]
	Benzene	78.11	Hydrocarbon, aromatic	[58, 61, 63, 66, 71, 72, 84]
	Octanal	128.21	Aldehyde	[61, 62, 64, 67, 73, 75, 85]
	Nonanal	142.24	Aldehyde	[61, 62, 64, 67, 73, 75, 86]
	Ethylbenzene	106.16	Hydrocarbon, aromatic	[61–63, 65, 71, 80, 82, 84]
	Hexanal	100.16	Aldehyde	[61, 64, 67, 73–75, 84, 87]

Given the substantial quantity of lung cancer markers, only those markers exhibiting more than five occurrences are presented, while the comprehensive list is available in Supplementary Table S4

VOCs are not definitively known, highlighting the necessity for additional research to elucidate the underlying biochemical mechanisms linked to the detection of these volatile substances.

In order to enhance the robustness of the findings, we conducted a systematic enumeration of VOC markers that have been consistently documented across multiple studies in the published articles (Supplementary Fig S3, Table S4). The findings indicate that 2-methylbuta-1,3-diene, propan-2-one, hexadecane, 1,3-dioxolan

2-one, 1,2,3-trimethylbenzene, and butan-2-one have been identified as a distinctive marker for gastric cancer by two independent research groups. Additionally, 11, 3, 1, and 1 analogous biomarkers have been detected in colon cancer, breast cancer, liver cancer, and malignant pleural mesothelioma (MPM), respectively. Numerous research studies focusing on lung cancer have identified a total of 69 distinct VOC markers, as reported by various research groups. Notably, propan-2-one, toluene, butan-2-one, propan-1-ol, benzene, ethylbenzene, and hexanal

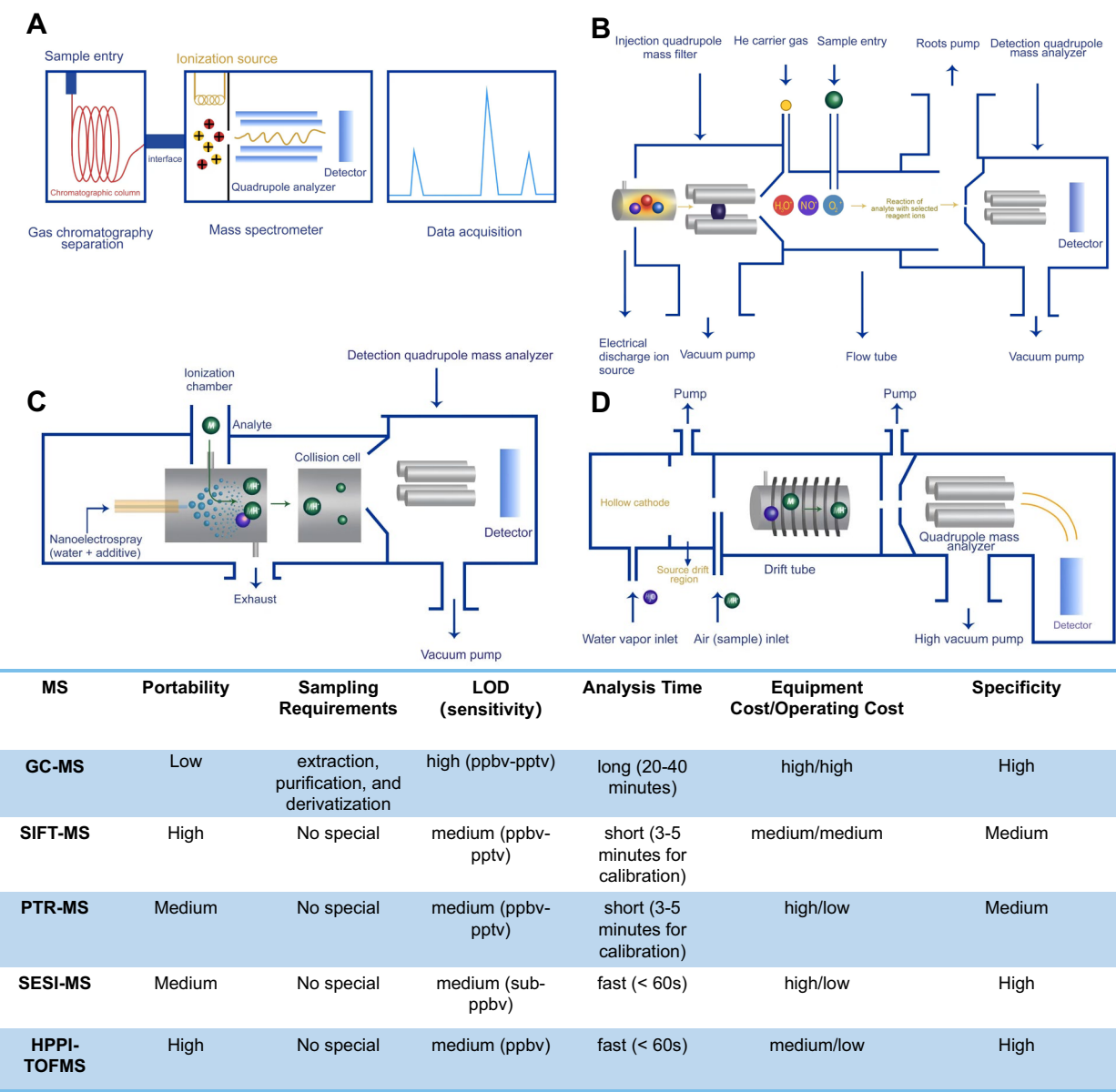


Fig. 2 Schematic representation of different techniques used for breath analysis: **A** GC-MS; **B** PTR-MS; **C** SIFT-MS; **D** SESI-MS. The table presented in the figure provides a comparative analysis of the four MS techniques alongside HPPI-TOFMS

are among the well-established markers of lung cancer, consistently documented in over eight scholarly publications. A comprehensive examination of existing literature within the realm of breath analysis reveals that certain VOCs have been proposed as potential biomarkers for various forms of cancer (Table 1). It has been observed that benzaldehyde, undecane, decanal, phenol, nonanal, ethanol, and toluene serve as markers in over six types of cancer, suggesting potential commonalities in biochemical pathways across various cancer types.

Mass spectrometry techniques for VOCs analysis

The extensive utilization of breath testing necessitates a technique possessing elevated sensitivity and precise selectivity capable of identifying a range of minute concentrations and types of VOCs in exhaled breath samples [88]. Given that the theoretical basis of VOCs in cancer diagnosis has not been thoroughly studied, the utilization of MS technology for qualitative and quantitative analysis presents a promising solution to address this gap [19]. The advancement of MS technology has led to the emergence of a variety of new technologies aimed at

addressing the limitations of previous generations. These innovations focus on improving pre-processing techniques, enhancing anti-interference capabilities, increasing analysis rates, and other related aspects (Fig. 2). In addition to the gold standard GC-MS, various other MS technologies are frequently employed, including Proton-transfer-reaction mass spectrometry (PTR-MS), Selected ion flow tube-mass spectrometry (SIFT-MS), Secondary Electrospray Ionization-Mass Spectrometry (SESI-MS), and High-Pressure Photon Ionization-time-of-flight mass spectrometry (HPPI-TOFMS), among others.

Gas chromatography-mass spectrometry (GC-MS)

GC-MS is a widely used analytical method that is highly regarded for its ability to analyze VOCs. It is considered the gold standard for this type of analysis and has the capability to detect more than 500 distinct compounds within a single exhaled breath sample [89]. The principle of GC-MS is shown schematically in Fig. 2A. Samples are introduced into an injection port and then, assisted by a carrier, typically an inert gas, they pass through a capillary column. At the end of the column, the compounds are ionized, traverse an electromagnetic field, and are filtered based on mass [90]. Subsequently, the MS analyzes the mass-to-charge ratio, offering a unique signature of the sample [91].

In the field of breath analysis, the process of pre-concentrating samples is a significant factor that impacts the subsequent processing methods and results. To overcome the constraints associated with traditional pre-concentration methods, the development of solid phase microextraction (SPME) has been introduced. In the context of GC-MS, pre-concentration techniques like SPME are commonly utilized to enhance the efficiency of the analysis. Nonetheless, these procedures are frequently time-consuming and may result in the presence of unknown contaminants [92]. Generally, the combination of consistent gas chromatography retention times and mass spectra enables the identification of metabolites using mass spectral libraries that can be used across different instruments, irrespective of the manufacturer [93]. The capability of GC-MS to detect and identify individual VOC molecules provides a distinct advantage in investigating underlying mechanisms. However, GC-MS is a costly, large, intricate instrument that is limited to laboratory settings and necessitates skilled operators for analysis. The duration required for sample analysis is extended, typically ranging from 30 to 60 min, thereby diminishing the efficiency of the analysis process [94]. Due to these factors, GC-MS is not conducive to real-time monitoring and is unsuitable for point-of-care or extensive disease screening applications [95]. A study conducted in 2021 involving 272 lung cancer patients and

261 control subjects investigated the differences in VOCs using GC-MS. The findings indicated a statistical marker for the early diagnosis of lung cancer, demonstrating an area under the receiver operating characteristic curve (AUC) exceeding 0.930, with a sensitivity of 90% and a specificity of 88% [86]. Owing to its enhanced stability and established technological framework, GC-MS is frequently employed in scientific research to validate the accuracy of alternative methodologies [82].

Proton-transfer-reaction mass spectrometry (PTR-MS)

PTR-MS has emerged as a powerful analytical technique for real-time detection and quantification of VOCs at trace concentrations ranging from parts-per-billion (ppbv) to parts-per-trillion (ppmv) levels [96, 97]. As shown in a simplified schematic in Fig. 2B, the generation of reagent ions H_3O^+ is facilitated through the discharge of water vapor within the hollow cathode ion source. Subsequently, a proton transfer reaction transpires from H_3O^+ to VOCs within the drift tube, resulting in the protonation of VOCs [98]. These reagent ions are generated through a hollow cathode ion source where water vapor undergoes electrical discharge. The resulting protonated molecules are then analyzed using high-resolution MS, typically employing time-of-flight (TOF) or hybrid quadrupole-TOF (QITOF) systems capable of achieving mass resolution exceeding 10,000 ($m/\Delta m$) [99, 100]. This configuration enables sensitive detection without requiring sample pre-concentration or pre-treatment.

A key advantage of PTR-MS is its ability to analyze samples in real-time with response times typically less than one second, making it particularly suitable for dynamic monitoring applications. In clinical applications, PTR-MS has shown remarkable potential for non-invasive disease diagnostics, particularly in cancer detection. Several studies have successfully differentiated between cancerous and non-cancerous cells across various cancer types including lung, esophageal, and cervical cancers. A notable 2022 study involving breath analysis of over 1,000 lung cancer patients demonstrated diagnostic capability with an AUC of 0.77 [101]. The technique's high sensitivity and specificity for VOC biomarkers, combined with its rapid analysis capability, position it as a promising tool for early disease detection. However, the current high instrument costs (typically exceeding \$500,000) and the need for specialized operation limit its widespread clinical adoption, keeping it primarily in the research domain. Ongoing developments aim to improve cost-effectiveness and robustness for potential future clinical implementation.

Selected ion flow tube-mass spectrometry (SIFT-MS)

SIFT-MS, developed by Smith and Spanel in 1995 [102], combines flow tube technology with chemical ionization and MS for real-time analysis of trace gases in ambient air and exhaled breath [55, 103]. As shown in a simplified schematic in Fig. 2C, the microwave discharge ionization source facilitates the generation of three main reagent ions (H_3O^+ , NO^+ , and O_2^+) from water vapor and air, which are selected by the first quadrupole mass analyzer to produce a current of given mass-to-charge ratio (m/z) ions [104]. The selected reagent ions subsequently undergo a reaction with the analyte within the flow tube [105], after which the ionized reagent ions and distinctive product ions can be identified through a downstream mass analyzer [10]. The Limit of Quantification (LOQ) values documented for SIFT-MS typically span from low ppbv to high pptv [106], encompassing a m/z range extending up to $m/z = 300$ [107].

Key advantages include real-time analysis capability without sample pre-concentration [104, 108]. Enabled by rapidly switchable reagent ions employing different ionization mechanisms. However, limitations include incomplete breath VOCs fingerprinting, difficulty in specific VOC identification based solely on m/z values [15], and inability to detect low-volatility compounds (requiring sample transfer to vacuum before ionization) [94]. In clinical applications, a 2021 study demonstrated SIFT-MS's potential in lung cancer diagnostics, analyzing 104 patients and 118 controls to identify 50 significant biomarkers, achieving 96% specificity and 88% sensitivity in validation [62]. While offering real-time analysis advantages, the technique's clinical adoption faces challenges due to these analytical limitations and the need for vacuum conditions. Current applications remain primarily research-focused, with potential for development in non-invasive disease diagnostics pending technological improvements.

Secondary electrospray ionization-mass spectrometry (SESI-MS)

SESI-MS was first observed by Hill in 1994 [109], and formally characterized in 2000 [110]. A schematic drawing of SESI-MS is depicted in Fig. 2D. A pure solvent with an additional component to enhance conductivity is utilized in the electrospray process to produce primary ions. Subsequently, the vapors of the analyte are directed towards the electrospray plume, resulting in ionization through proton-transfer reactions. This process predominantly yields molecular ions [111, 112], with recent evidence suggesting gas-phase ion-molecule reactions (involving charged water clusters) play a more significant role than previously assumed gas-droplet interactions [113].

The technique achieves sensitivity of pptv-level, with exceptional sensitivity (sub-ppqv levels) for polar compounds [114, 115], and can detect low-volatility, biologically relevant molecules [116]. In addition to the notable efficiency and gentle ionization characteristics of SESI, its primary advantage lies in its modularity. Its key advantages include atmospheric pressure ionization (unlike PTR-MS/SIFT-MS), enabling: (1) enhanced ionization efficiency through pressure modulation [117], and (2) direct coupling with high-resolution mass spectrometers ($> 100,000$ resolution, e.g., Orbitrap) [94]. SESI-MS is frequently combined with high-performance gas chromatography (HPLC) for breath condensate analysis. SESI-MS is frequently combined with HPLC for breath condensate analysis. Despite these advantages, absolute gas-phase quantification remains challenging [10]. Clinical applications leverage SESI-MS's high sensitivity for breath analysis, though its current research use reflects both its analytical potential and quantification limitations. The technique's modular design and compatibility with existing instrumentation suggest promising avenues for future diagnostic development.

High-pressure photon ionization-time-of-flight mass spectrometry (HPPI-TOFMS)

HPPI-TOFMS, introduced in 2016 [118], addresses analytical challenges in exhaled breath analysis by combining high-pressure photoionization (100–500 mbar) with time-of-flight MS. The technique employs UV photons to generate molecular ions with minimal fragmentation, followed by high-resolution separation ($R > 3000$) based on mass-to-charge ratios. Key advantages include soft ionization preserving molecular structure, high sensitivity (ppbv–pptv range), rapid analysis (< 1 s/spectrum), and tolerance to water vapor interference [118]. These features make it particularly suitable for real-time analysis of complex gas mixtures.

In recent years, HPPI-TOFMS has been extensively investigated for its application in the diagnosis of cancer through exhaled breath analysis. In a 5047-patient multicenter study, HPPI-TOFMS identified 10 VOC markers enabling non-invasive BC detection. The BreathBC-Plus model (VOCs + risk factors) achieved 94% accuracy ($\text{AUC} = 0.94$), outperforming mammography. Detection rates reached 97% for ductal carcinoma in situ (DCIS) and 85–100% across cancer stages, with 88% specificity—demonstrating HPPI's clinical potential for precision screening [33]. However, compound identification remains restricted to m/z information without fragmentation patterns [119], and the technique requires further validation of potential biomarkers. Instrument costs remain high, comparable to other high-end MS systems, which may limit clinical adoption.

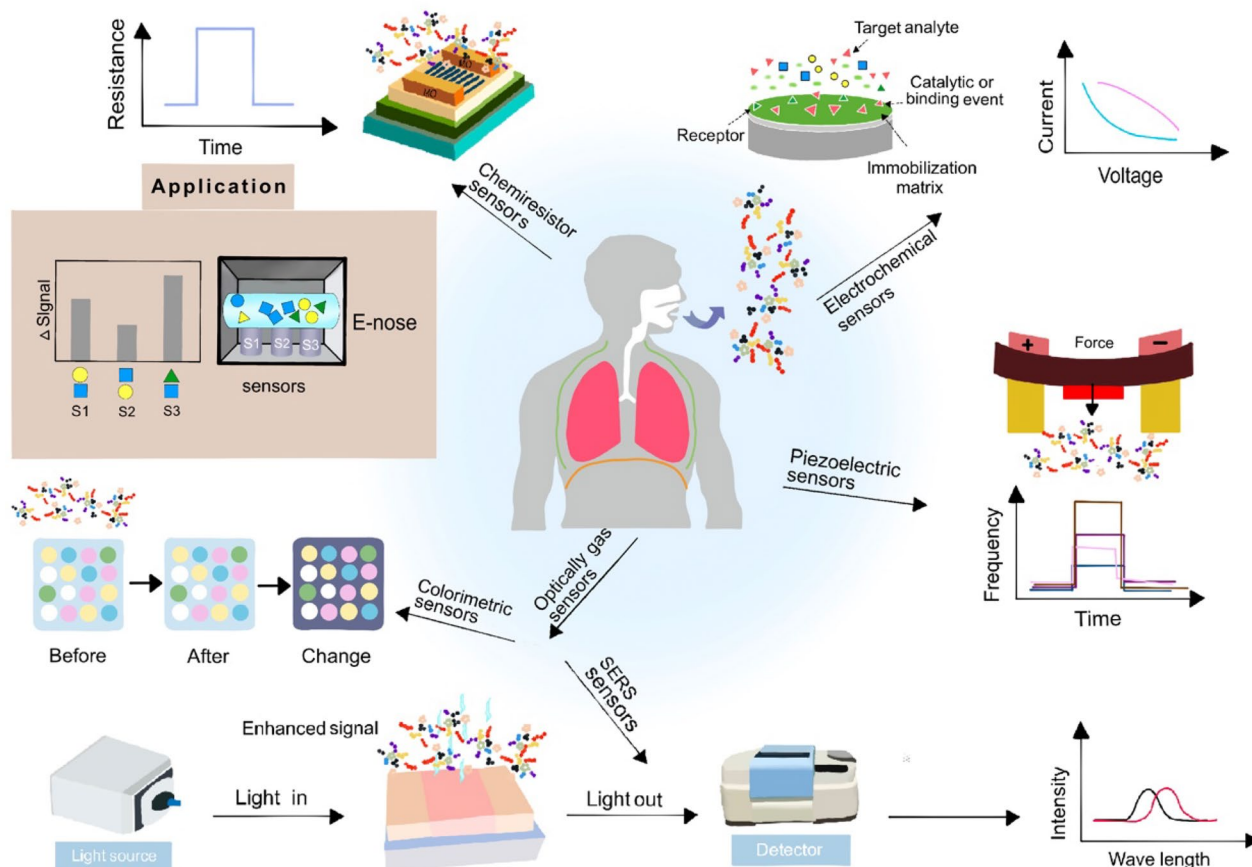


Fig. 3 Sensors as a volatolomic analytical platform: simplified working mechanisms. These sensors hold potential for application in future diagnostic devices that are based on volatolomic analysis within the medical field. Clockwise: chemiresistor sensors, electrochemical sensors, piezoelectric sensors, and optically gas sensors

Other MS techniques

Comprehensive two-dimensional gas chromatography-mass spectrometry (GC × GC-MS) represents a significant advancement in breath analysis, employing two orthogonal separation columns coupled with a modulator to achieve superior compound separation. This configuration provides enhanced peak capacity, resolution, and detection sensitivity compared to conventional GC-MS, making it particularly valuable for analyzing complex breath VOC profiles [120]. Fourier-transform ion cyclotron resonance mass spectrometry (FT-ICR-MS) offers unparalleled analytical performance with mass resolution exceeding 300,000 and sub-ppm mass accuracy, enabling precise identification of volatile organic compounds [121]. However, the technique's clinical implementation faces challenges due to its substantial costs. The system requires specialized components including a high-performance computing platform for Fourier transform analysis and a sophisticated ion trapping chamber, factors that

currently limit its widespread clinical adoption despite its exceptional analytical capabilities [122].

Sensors for VOCs detection

A wide range of sensor types has been developed over time, with numerous applications in the field of biomedicine. Various nano-sensors have been created, including chemiresistor sensors, electrochemical sensors, piezoelectric sensors, and optical gas sensors, such as colorimetric sensors, surface-enhanced Raman scattering (SERS) sensors, and optical fiber sensors. Among these, electronic noses (E-noses) are representative products for portable applications (Fig. 3). These sensors show significant promise in detecting VOCs and have potential applications in breath analysis due to their exceptional sensitivity and responsiveness. To further enhance sensor performance, a variety of materials have been explored [123]. The utilization of nanotechnology in chemosensors has seen a rise in recent years, leading to a proliferation of related research articles. Various nanomaterials, such

as nanoparticles, nanowires (NWs), and carbon nanotubes (CNTs), have been employed as sensing elements for VOCs. The diminutive scale of these components offers several advantages, including a high surface-to-volume ratio and distinctive chemical, optical, and electrical properties. The expanded surface area of nanomaterials results in highly reactive interfaces, thereby enhancing sensitivity and reducing response and recovery times [124]. Moreover, the nanoscale dimensions render nanomaterials responsive to localized entities of comparable size, ranging from minute molecules to substantial macromolecules.

Chemiresistor sensor

Chemiresistor sensors represent a promising category of gas sensors, characterized by their straightforward design and the capacity for accurate and precise measurements. Typically, these sensors consist of a semiconductor or metallic sensing layer that bridges two sets of interdigitated electrodes. Upon exposure to gas, the resistance of the sensor alters, while the potential difference between the electrodes remains stable [125]. The quantification of analytes can be accomplished by calculating the variations in resistance or current [126]. A range of nanomaterial-based sensing films is employed for the detection of VOCs, including nanotubes (NTs), graphene, CNTs, NWs, metal oxide semiconductors (MOS), metallic nanoparticles (MNPs), and hybrid nanomaterials. Due to their cost-effectiveness, ease of production, portability, and high sensitivity to gases [127], MOS, including In_2O_3 , and SnO_2 , along with transition metal oxides such as ZnO , Cr_2O_3 , WO_3 , and NiO , are regarded as optimal candidates for resistive gas sensing applications. Carbon-based nanomaterials, particularly graphene and its derivatives, including graphene oxide, have demonstrated significant potential in the detection of gases and are regarded as promising materials for gas sensitivity applications [16]. This efficacy can be attributed to their remarkable specific surface area, exceptional thermal stability, and high carrier mobility. Sensors utilizing single-walled carbon nanotubes (SWCNTs) have exhibited superior responsiveness in comparison to those based on multi-walled carbon nanotubes (MWCNTs), which have indicated their potential applicability in electrochemical sensors [16]. Nevertheless, a significant limitation encountered in the deployment of CNT-based sensors is their diminished chemical selectivity [128]. The high working temperature of oxides and less selectivity of CNTs inhibits the purpose of having a high-performance device.

Electrochemical sensors

Electrochemical sensors represent a distinct category of sensors utilized for the detection of VOCs. The electrochemical sensors are composed of two or three electrodes and an electrolyte solution, forming a closed circuit. The concentration of VOCs measured by these sensors is directly proportional to the resulting current generated. This particular type of sensor functions within a closed circuit, enabling the connection of multiple cells in series, each selectively responsive to a specific compound. A significant advantage of this sensor configuration lies in its capacity to construct intricate structures, facilitating the concurrent analysis of various VOCs and enhancing diagnostic capabilities for cancers [129]. Moreover, electrochemical sensors are increasingly favored among researchers in the field of breath analysis due to their well-established operating principles, high accuracy, low detection limits, broad detection range, biocompatibility, potential for miniaturization, low power consumption, and cost-effectiveness. However, these sensors exhibit certain limitations, including prolonged response times and an inability to detect long-chain VOCs. While enzyme-based electrochemical sensors demonstrate high selectivity, non-enzymatic sensors are prone to cross-sensitivities arising from breath moisture and other competing VOCs and gases. Furthermore, electrochemical cells utilizing aqueous electrolytes results in a diffusion-controlled process, leading to extended response times. Various techniques have been explored to address these challenges, with "fuel cell" technology employing solid electrolytes emerging as the most effective solution [130].

Piezoelectric sensors

The application of mechanical stress to the surface of a sensitive layer within a piezoelectric material results in the generation of an electrical voltage [131]. This phenomenon serves as the foundational principle for piezoelectric sensors, with the Quartz crystal microbalance (QCM) and Surface acoustic wave sensor (SAW) serving as prominent examples [132].

When a voltage is applied to the upper electrode of the QCM, an acoustic wave is generated and transmitted through the quartz piezoelectric substrate. This acoustic wave is subsequently converted back into an electrical signal, which is quantified in terms of resonance frequency. For the purpose of sensing, the QCM is coated with a specific sensing layer designed to adsorb the target analyte gas. The interaction between the sensing layer and the analyte gas results in alterations in both the amplitude and velocity of the acoustic wave, attributable to variations in mass [133, 134]. Hence the shift of the resonance frequency will be observed due to the mass variations [135]. The adsorption of gases onto the

QCM surface can be modulated through the application of different coatings, including polymers, metal oxides, and nanomaterials [130]. The application of chemical layers that exhibit sensitivity to acetone, including materials such as ZnO, WO₃, SnO₂, and In₂O₃, can significantly improve the sensitivity of these sensors [16]. The capability of QCM sensors to conduct real-time monitoring of target analytes [16], coupled with its high sensitivity and low power consumption [136], constitutes key attributes that enhance its utility.

SAW sensor works slightly different from QCM sensor in terms of the propagation of the acoustic wave. The acoustic wave generated by a SAW sensor propagates along the surface of the substrate. When a radio frequency voltage is applied across the interdigitated transducer (IDT), it induces deformations in the piezoelectric substrate, resulting in the formation of a Rayleigh surface wave that is confined to a single acoustic wavelength on the surface [137]. As this wave traverses the gap between two arrays of IDTs and reaches the receiver IDTs, the mechanical energy is converted back into radio frequency, producing an output signal [134]. Any changes in mass due to a coating or the adsorption of vapor on the surface lead to a frequency shift, thereby facilitating the application of this technology in vapor monitoring. Similar to QCM, each individual material necessitates modification to enhance the selectivity and sensitivity of the sensor. Additional research into the properties of each material and the analyte gases is essential to further improve the application of acoustic wave sensors for gas sensing purposes [138].

Optically gas sensors

Colorimetric sensors

Colorimetric sensors are utilized in optical techniques, facilitating cross-reactive detection due to their capability to simultaneously identify multiple molecules. The underlying principles of these sensors are predicated on the optical alterations detected in colorants. A diverse array of dyes may be employed for the identification of various compounds, contingent upon their specific affinity for the analyte in question. Sensors of this nature are typically characterized by their reliability and user-friendliness. Consequently, they are well-suited for development as wearable devices [139]. Intermolecular forces play a significant role in determining the efficacy of the sensor. However, certain critical factors, such as humidity, can influence the ultimate results, given that these interactions are typically characterized by their relative weakness. As a result, the primary characteristics of the sensor, including its responsiveness, sensitivity, and detection limits, are diminished [140]. Consequently, it is evident that the stabilization of a sensor necessitates

robust intermolecular interactions. It is essential to design sensors utilizing nanomaterials that establish stronger connections with VOCs than those provided by Van der Waals forces.

Surface enhanced Raman scattering (SERS) sensors

SERS is an optical sensing technique that has a wide range of applications. It has been employed as a gas analytical method that is tailored to detect specific VOCs or a group of such compounds, utilizing the principles of Surface Plasmon Resonance (SPR) [125]. The surface of the SERS sensor is typically coated with a layer of functionalized molecules that exhibit high selectivity and affinity for particular VOCs, facilitating their capture. The application of laser irradiation to the SERS-active surface generates an enhanced Raman scattering signal. The subsequent analysis of the Raman spectra allows for the differentiation of VOCs present in various respiratory samples. The initial challenge pertains to the absorption difficulties encountered on the surfaces of materials, which contribute to the low detection rates, as gas molecules exhibit high velocities. Additionally, Raman signals may be undetectable in certain gases due to their low concentrations [141]. These challenges can be addressed by altering the morphology of the nanomaterials employed or by integrating nanoparticles with a high affinity for the analyte molecules.

Portable applications of VOCs sensors

For portable applications of VOCs sensors, E-nose is a kind of mature device that mimics the human sense of smell by detecting and analyzing volatile compounds in gases, which has been widely used in medical diagnostics and environmental monitoring. It typically consists of an array of sensors, a signal processing unit, and a pattern recognition system. The fundamental component of the E-nose consists of a collection of sensors, such as metal oxide semiconductors, carbon-based nanomaterials, and piezoelectric materials, which exhibit responsiveness to particular VOC gas molecules [142]. These technologies are grounded in distinct principles and employ various sensing materials and array configurations [143]. The E-nose is designed to replicate the functionality of the human olfactory system, enabling the identification of gases according to their chemical composition. To enhance its performance, it is essential to utilize diverse functionalizations [144]. The E-nose can be categorized into two types based on the number of functional classifications it contains: homogeneous and heterogeneous E-nose.

The advancement of numerous biomedical applications utilizing electronic-nose technologies has encompassed over a dozen distinct commercially available E-nose

	Detecting method	MS technology	Sensors	
P (Participants)	· Age, Sex, Race, Country/Region · Smoking or alcohol consumption, fasting or no fasting · Current medication, Drug of abuse · Stage of the index cancer · Comorbidities · Cardiac and pulmonary functions			
I (Index test) VOC-based breath test	Breath Sampling			
	Preparation	· Prior fasting · Prior drink intake	· Forbid prior exercise · Forbid prior smoking	· Patient education · Prior breath practice
	Collection	Approach	Direct OR Indirect	Direct
		Timing	Morning OR Anytime of day	
		Device	Disposable or needs washing? With inlet device?	Not involved
		Fraction	Mixed OR Alveolar	Mixed
		Maneuver	Through Nose OR Mouth	
		Duration	Single breath? Fixed-time?	
	Sample storage	· On adsorbent materials or in bags · Storage Temperature · Comparison before and after storage	Not involved	
	Sample pre-treatment	· Adsorption · Desiccation	Not involved	
	Sample transfer	· Turn-over time · Economic cost	Not involved	
	VOC measurement			
	Analysis platform	Untargeted analysis molecule identification-based	Single sensor	
		Targeted analysis molecule panel-based	E-nose	
	Analysis calibration	Validation compared to GC-MS		
	Statistical analysis			
	Internal validation	Split-half, (X-fold) Cross-validation, X-fold analyses, Bootstrapping		
	External validation	Multi center/Single center		
C (Comparator test)	Biopsy, etc.	Limited time interval Double-blinded		
O (Outcomes)	For test accuracy	Sensitivity, Specificity, PPV, NPV, AUC		
	For care with test	All-cause death, Possible fatal adverse events		
S (Study design)	Study design	Cross-sectional case-control studies		
	Study population	Population with realistic disease prevalence OR not		

Fig. 4 Proposed standardized study framework for future studies investigating the role of VOCs in cancer diagnosis. The factors that would influence the quality of the VOCs-based diagnostic studies is exhibited in a PICOS framework. These factors are summarized from the methodological review and quality assessment of this systemic review. The proposed framework could guide the conduction of future breath tests in cancer studies

instruments. To date, the predominant majority of sensors utilized in medical contexts, such as chemiresistors, SERS, QMB, SAW, electrochemical sensors and colorimetric sensors, constitute a limited variety of the E-nose technology types that are currently accessible [145]. The majority of biomedical electronic nose applications have employed MOS sensor arrays, which consist of a varying number of sensors. The reliance on MOS sensors is somewhat unexpected, as these sensors typically necessitate elevated operating temperatures and exhibit significant power consumption for detection purposes. Moreover, MOS sensors exhibit heightened sensitivity to humidity, are vulnerable to poisoning from sulfur and weak acids, and possess constraints regarding the types of sensor coatings that can be applied [146]. QCM and SAW sensors, conversely, present several advantages, including minimal power consumption, the capability for real-time detection, and the straightforward modification of their sensing surfaces, which allows for the adjustment of their sensitivity and selectivity. A variety of nanomaterials, including graphene, CNTs, MNPs, MOS, metal monolayer-capped nanoparticles (MCNPs), and polymers, have been utilized as constituents in E-noses. The modification or combination of these nanomaterials facilitates the creation of hybrid materials, which expands their potential applications, enhances their cross-selectivity and sensitivity, and enables precise pattern recognition [142].

Workflow of VOCs detection

A qualitative review of the current research workflow was conducted and subsequently summarized. The difference in detection process between MS and sensor technology is shown in Fig. 4. The established protocol for breath collection should encompass the establishment of a study cohort, information collection, patients' preparation, the sample collection, pretreatment procedures, the sample transfer, analysis of VOCs, and subsequent statistical analysis. The detection of exhaled air can be categorized into two primary methodologies: direct analysis and indirect analysis. Direct analysis consolidates the processes of sample collection, analysis, and model construction within a single apparatus, thereby facilitating the direct output of results and streamlining the overall procedure. Conversely, indirect analysis encompasses a more intricate process, which may involve various stages such as sample collection, transportation, pretreatment, and the development of algorithms. It is noteworthy that the majority of MS techniques employed in research predominantly rely on indirect analysis. Owing to the distinctive principles underlying sensor pattern recognition, direct analysis is frequently employed in the context of utilizing sensors for the detection of exhaled air. The

following aspects are summarized: patients' preparation for sampling, the maneuver of breath sampling, method of breath collection, transfer and turnover time, analytical platform, and the methodology of diagnostic model construction.

Patients' preparation for sampling

Patients' preparation for sampling includes fasting, resting, and abstaining from smoking, brushing teeth, and using perfume. The purpose of these preparations when collecting exhaled breath for cancer diagnosis is to avoid interference from external substances that may affect the accuracy of the test results. These interfering factors could potentially mask or disrupt the detection of cancer biomarkers in exhaled breath, thus impacting the accuracy of the diagnosis [89]. Fasting helps to avoid residue or volatile compounds from food affecting the exhaled breath sample, most included studies require fasting for 4–10 h (Supplementary Table S5). Resting reduces the impact of physical activity while abstaining from smoking helps to prevent the impact of compounds from tobacco on the exhaled breath sample. Not brushing teeth helps to avoid the influence of toothpaste, mouthwash, and other oral care products on the exhaled breath sample, and refraining from using perfume or related products serves the same purpose. In summary, these restrictions aim to ensure the purity of the exhaled breath sample and enhance the accuracy and reliability of cancer diagnosis. In order to evaluate the influence of patient' preparation on the accuracy of the test, subgroup analysis (fasting/no fasting) was performed in the results of MS detection (Supplementary Fig S4). The findings indicated that there was no statistically significant difference in specificity and sensitivity between the two subgroups.

Breath sampling maneuver

Breath sampling maneuver includes collecting gas and exhaled breath condensate (EBC). Exhaled breath can be divided into three stages [147]. The first stage contains gas from the oral cavity and trachea ("dead space" stage). The next stage is the "mixed exhaled" phase, where gas from the lungs and lower respiratory tract mixes with the dead space sample. The final stage is the "end-tidal" phase, which best reflects the gas from the lung alveoli, concerning cancer-related metabolism. The first two phases should not be discarded as they may contain biomarkers most relevant to upper digestive tract cancers. Most breath analysis focus on collecting the alveolar gas. This can be achieved by real-time monitoring of carbon dioxide levels during sampling [148]. Carbon dioxide is solely derived from alveolar gas exchange and serves as a marker for the end-tidal phase.

Sampling maneuver can also be classified as nasal, and oral route, which was the mainstream maneuver. Contaminant VOCs from the Gastrointestinal (GI) tract should not be ignored. The duration of sample collection depends on the technique used, which could be a single exhalation (in seconds), or in minutes. In bag or online sampling, multiple samples can be collected continuously to reduce inter-individual variability in breath levels. There is currently no universal protocol for breath gas sampling. Markar et al. specified the identified unavoidable contaminant VOCs in ambient room air, including butanol and acrolein [149]. Thus, ambient room air should be collected to ensure no contamination interfering with breath samples.

Beyond the gaseous phase, breath can also be collected and liquefied as EBC. EBC consists of condensed water vapor (> 99%) and of aerosolized droplets of airway lining fluid. These respiratory droplets contain non-volatile, hydrophobic/water-soluble molecules, which are also absorbed by the condensed water vapor during EBC collection [150]. Thus, non-volatile compounds (such as organic acids, nucleic acids, amino acids, urea, peptides, proteins and macromolecules) in EBC compensate for volatile biomarkers [150]. EBC is typically collected at temperatures below 4 °C. The condensation temperature can affect the concentration of different biomarkers, so reporting the condensation temperature is crucial for process standardization and comparison across studies [151]. Various commercial EBC collection devices are available, such as EcoScreen Turbo [152], Turbo-DECCS [153], Rtube [154]. It is evident that the duration of sample collection directly affects the final volume [155]. Most published studies suggest collecting EBC samples for 10 to 30 min [156], and the approximate volume of EBC is 1 to 3 mL. The optimal temperature for storing EBC samples is below - 70 °C [157].

In order to evaluate the influence of the "dead space" air on the accuracy of the test, subgroup analysis (alveolar/mixed) was performed in the results of MS detection (Supplementary Fig S4). The results demonstrated that there was no statistically significant difference in specificity and sensitivity between the two subgroups.

Breath collection container/device

The ideal collection container should be cost-effective, durable, and chemically inert, and airtight. In most studies, collection bags were used as temporary storage, with a mouthpiece, syringe, inlet, or fitting to exhale through. Polymer bags, particularly Tedlar® bags (E.I. du Pont de Nemours and Company, Wilmington, DE, USA), are the most used containers for breath collection [158]. Other polymer bags such as Mylar and aluminum bags are also used. The optimal temperature for storing bags is 37 °C to

avoid condensation on the bag surface. Tedlar® bags have been proven to be stable for storage for up to 6 h [159].

For long-term storage, thermal desorption (TD) tubes and glass vial are used because they are less permeable [42, 160]. Breath sampling devices such as ReCIVA® breath sample system, Bio-VOC™ Sampler, and breath collection apparatus collect fractionated breath directly to the TD tubes/glass vials without a bag collection procedure [54, 161]. Studies have shown that these tubes can be stored for up to 2 weeks at - 80 °C prior to analysis [162, 163]. Among nearly 600 compounds studied, a significant concentration change was observed for a considerable portion after 6 months of storage [164].

Turnover time refers to the time interval from collection to MS analyses. It concerns the applicability of VOCs-based diagnosis in the clinical setting and the diagnostic test stability. The stability of target VOCs is associated with temperature, the permeability of the collection bag, and transfer time. Although included studies did not specify the turnover time, it is a crucial evaluation index worth reporting in future studies.

Pre-concentration method

To detect VOCs present in breath at parts per million by volume (ppmv) or lower concentrations, preconcentration is typically required. With certain collection methods, there may be dilution of the VOCs, especially when combining exhaled samples. Common preconcentration methods include the use of TD tubes containing adsorbents, SPME, and Needle Trap Devices (NTDs).

TD tubes are commonly used for the preconcentration of VOCs and account for almost half of the published preconcentration methods to date. Commonly used adsorbents include Tenax TA&GR, Carbograph 5TD, and Carboxen. Strong adsorbents like Carboxen are suitable for capturing highly volatile organic compounds (~ C2-C4), while Tenax adsorbents can capture lower volatility VOCs (~ C7-C15) present in breath [165]. Factors to consider include whether to use a single-bed adsorbent or a multi-bed adsorbent, which will depend on the analytes of interest and the quantity of adsorbents used.

SPME is a preconcentration technique developed by Pawliszyn et al. and has been applied in breath analysis [166]. It involves exposing a coated fused silica fiber to the headspace of the sample. The coating materials include polydimethylsiloxane and polyacrylat, among others [167]. This method has been shown to possess sensitivity comparable to TD tubes but may be limited by coating amount and fiber thickness [167].

NTD is also used to capture analytes from exhaled breath. In simple terms, an adsorbent material is confined within a needle-like device, and breath is "pulled" through the needle to capture VOCs. This method is

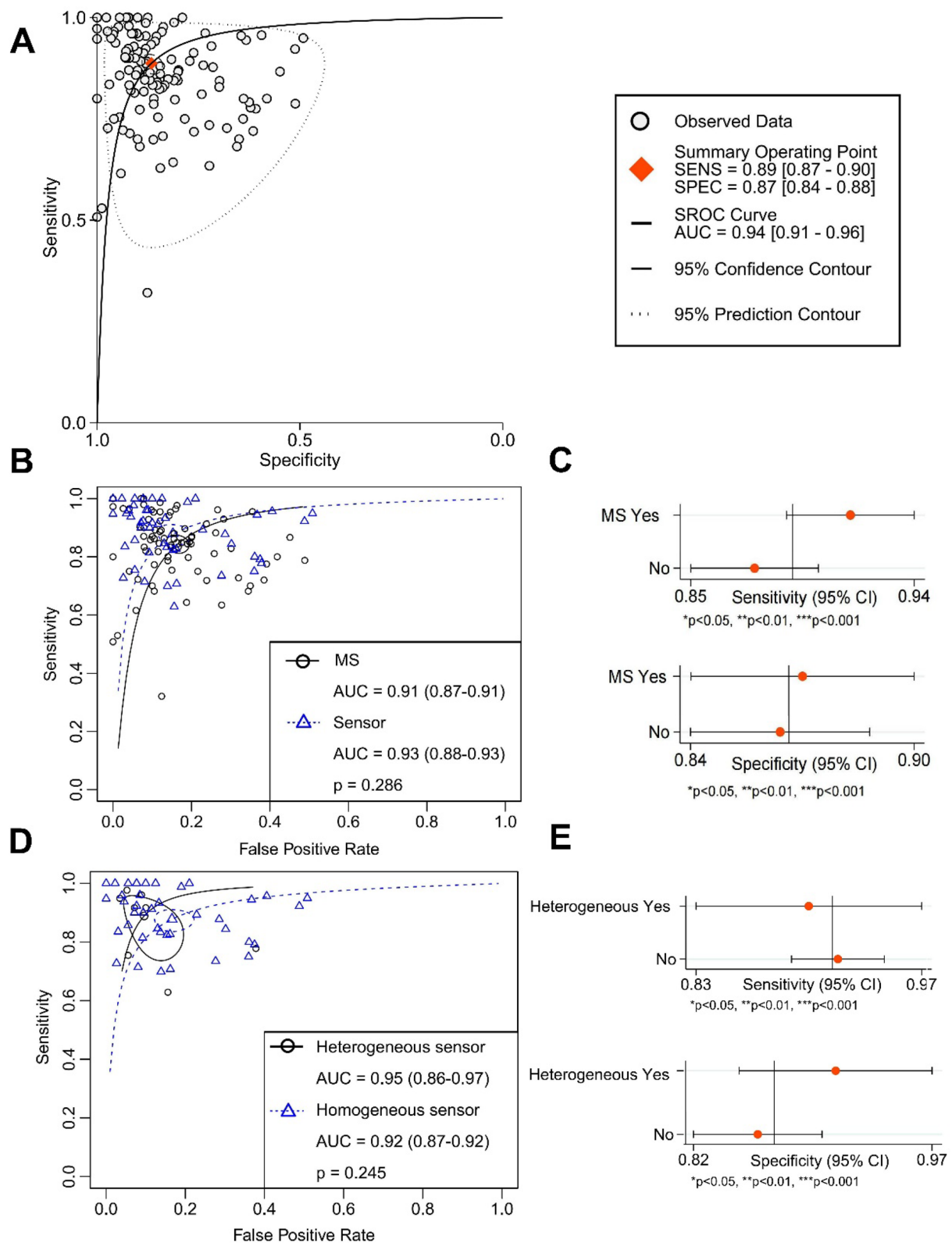


Fig. 5 Diagnostic value of VOCs-based breath test and Comparison of diagnostic efficiency of different detection techniques. **A** Summary receiver operating characteristic (sROC) curve. **B** sROC Curves for VOCs-based breath test of MS technologies and sensor detection. **C** Subgroup analysis for VOCs-based breath test of MS technologies and sensor detection. **D** sROC Curves for VOCs-based breath test of heterogeneous and homogeneous sensor detection. **E** Subgroup analysis for VOCs-based breath test of heterogeneous and homogeneous sensor detection

Table 2 The diagnostic efficacy of different detection methods in different types of cancer

Cancer type	GC-MS	PTR-MS	SIFT-MS	HPPI-TOFMS	Sensor
Lung	0.91	0.71 [*]	0.96 [#]	0.91	0.93
Gastroesophageal	0.91	0.93	0.94	0.98 [#]	0.95
Breast	0.83	-	0.72 [#]	0.94 [*]	0.92 (p = 0.069)
Head and neck	0.95	-	0.90	-	0.92

All AUC values were analyzed using p-tests to evaluate the diagnostic efficacy of GC-MS for this specific type of cancer, with significance levels set at *p < 0.05. In instances where the quantity of articles is insufficient, the p-test cannot be utilized. Such cases are denoted with a hashtag (#)

designed to incorporate the best features of SPME and TD tubes, aiming to improve detection limits, reduce sampling time and volume, as well as improve stability and reproducibility [168]. Like TD tubes, sensitivity can be enhanced by increasing the sampling volume. It has been reported that samples obtained using SPME [166] and NTD [169] remain stable for several hours before significant losses are observed.

Diagnostic model construction

Constructing a diagnostic model for VOCs-based breath test included two aspects: cancer-associated VOC biomarker selection and diagnostic modeling (Supplementary Table S1). Partial least squares discriminant analysis (PLS-DA), Mann-Whitney U test, and Kruskal-Wallis test were most adopted to select markers. In the realm of diagnostic modeling, various techniques such as logistic regression analysis (LRA), random forest (RF), support vector machine (SVM) and weighted digital analysis (WDA). Some studies lack a validation group and used leave-one-out cross-validation (LOOCV).

Results

Accuracy of VOCs-based breath tests in diagnostic studies
In all the included studies, the AUCs of VOCs-based breath tests for cancer diagnosis ranged from 0.65 to 1.00. The sensitivity and specificity ranged from 32 to 100%, and 56% to 100%, respectively (Supplementary Table S1). Meta-analysis demonstrated mean AUC of receiver operating characteristic (ROC) curve of 0.94 (95% CI 0.91-0.96), sensitivity of 89% (95% CI 87%-90%), and specificity of 87% (95% CI 84%-88%) (Fig. 5A), demonstrating high potency of cancers diagnosis. Considering the nonnegligible heterogeneity, we listed the diagnostic efficacy of different analytical platforms in the same cancer type, and studies with the largest sample size were chosen (Supplementary Table S6). No observable advantages or disadvantages were identified among the various detection methods. We conducted a comparative analysis of the diagnostic efficacy of various detection methods across different cancer types to inform future research trajectories (Table 2). In the context of lung cancer diagnosis, GC-MS exhibited a markedly higher

diagnostic efficacy, achieving an AUC of 0.91, which is significantly superior to the 0.71 AUC recorded for PTR-MS (p = 0.016). The diagnostic performance of HPPI-TOFMS and various sensors was found to be comparable to that of GC-MS. Conversely, in the cases of gastric and esophageal cancers, no statistically significant differences were observed in the diagnostic efficacy of PTR-MS, SIFT-MS, and sensors when compared to GC-MS, with all aforementioned methods yielding AUC values exceeding 0.9. Notably, in breast cancer, the novel technology HPPI-TOFMS demonstrated commendable diagnostic efficacy, significantly surpassing that of GC-MS (p = 0.018). For head and neck cancer, the diagnostic efficacy of the SIFT-MS sensor was found to be comparable to that of GC-MS, while other detection methods have yet to be implemented. The detection of exhaled air demonstrates promising advantages over existing methodologies for cancer diagnosis (Supplementary Table S7).

Performance comparison of MS and sensor technology

In the preceding sections, we have reviewed the principle that MS and sensors are capable of differentiating cancer patients from healthy individuals through the detection of VOCs in exhaled breath. Nonetheless, there is currently a lack of empirical evidence to substantiate the diagnostic efficacy of MS and sensors in this context. Consequently, we consolidated the two analytical platforms and conducted a comparative assessment of the diagnostic efficacy of both (Fig. 6 and 7).

In the MS group, the AUCs of VOCs-based breath tests for cancer diagnosis ranged from 0.67 to 1.00. The sensitivity and specificity ranged from 32% to 100%, and 56% to 100%, respectively (Supplementary Table S1). Meta-analysis demonstrated mean AUC of ROC curve of 0.91 (95% CI 0.87-0.91) (Fig. 5B), demonstrating high potency of MS detection. Similar results were observed in the meta-analysis of the sensor group, with AUCs ranging from 0.74 to 0.99. The sensitivity and specificity ranged from 63% to 100%, and 49% to 100%, respectively (Supplementary Table S2). Meta-analysis demonstrated mean AUC of ROC curve of 0.93 (95% CI 0.88-0.93) (Fig. 5B). There was no statistical difference in the AUCs for MS

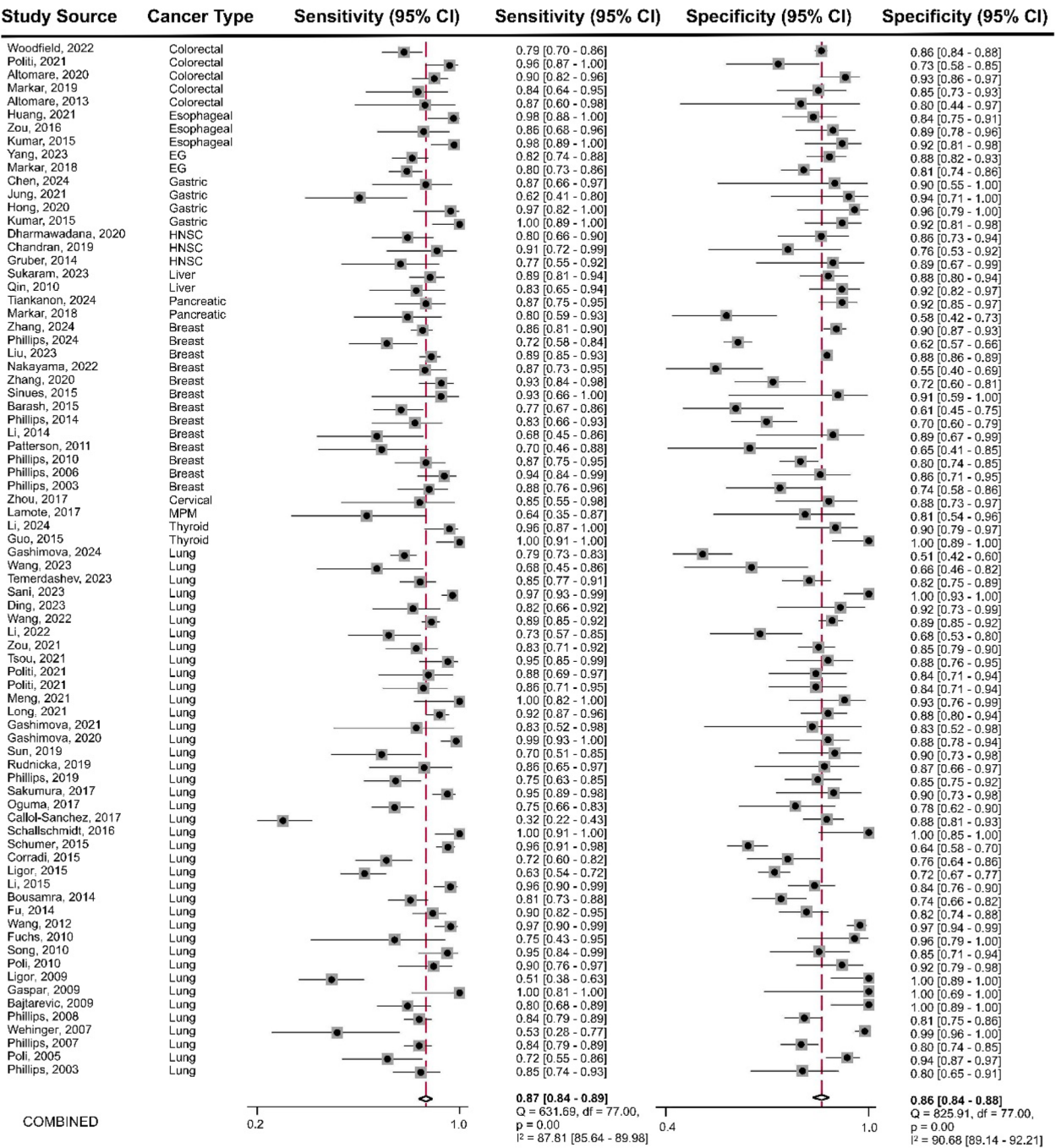


Fig. 6 Forest Plot of Pooled Sensitivity and Specificity Analysis of MS technologies. The width of the horizontal line represents the 95% CI of the individual study. The diamond represents the pooled RR and 95% CI

and sensors ($p = 0.286$). Subgroup analysis showed no statistical difference in specificity and sensitivity between the two groups (Fig. 5C).

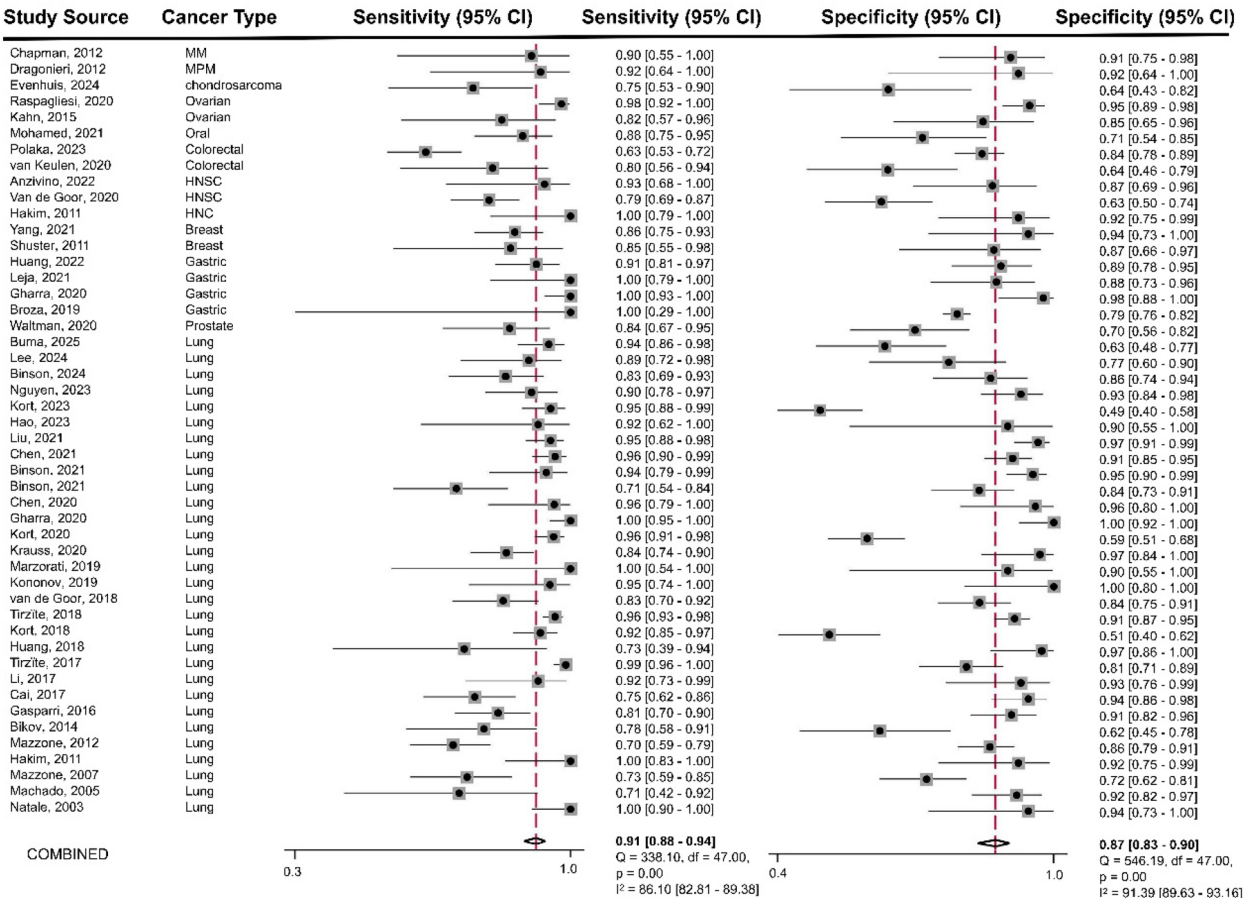


Fig. 7 Forest Plot of Pooled Sensitivity and Specificity Analysis of sensor detection. The width of the horizontal line represents the 95% CI of the individual study. The diamond represents the pooled RR and 95% CI

Performance comparison of heterogeneous and homogeneous sensors

The prevalent electronic noses available in the current market typically consist of multiple types of sensors. However, the extent to which this amalgamation confers a statistical diagnostic benefit remains uncertain. Therefore, we also compared the performance between heterogeneous and homogeneous sensors (including homogeneous E-noses), and conducted a comparative assessment of the diagnostic efficacy of both.

In the homogeneous-sensor group, the AUCs of VOCs-based breath tests for cancer diagnosis ranged from 0.69 to 0.99. The sensitivity and specificity ranged from 70% to 100%, and 49% to 100%, respectively (Supplementary Table S1). Meta-analysis demonstrated mean AUC of ROC curve of 0.92 (95% CI 0.87-0.92) (Fig. 5D). Similar results were observed in the meta-analysis of the heterogeneous-sensor group. The sensitivity and specificity ranged from 63% to 98%, and 63% to 97%, respectively (Supplementary Table S1). Meta-analysis demonstrated mean AUC of ROC curve of 0.95 (95% CI

0.86-0.97) (Fig. 5D). There was no statistical difference in the AUCs for heterogeneous and homogeneous sensors ($p = 0.245$). Subgroup analysis showed no statistical difference in specificity and sensitivity between the two groups (Fig. 5E).

VOCs detection and tumor stage

Few studies conducted well-established subgroup analysis according to tumor stage. Specifically, twenty-three studies undertook preliminary explorations with small sample sizes, resulting in low statistical reliability, and did not yield specific conclusions [28, 33, 36, 37, 40, 42, 46, 54, 170–176]. Such preliminary explorations included a comparison of specific VOC indicators between different stages [28, 37, 40, 46, 70, 83, 86, 171, 172, 174, 175, 177–179], diagnostic accuracy for distinguishing early and late-stage cancers [86, 170, 175, 176], and comparison of diagnostic accuracy for diagnosing different stages of cancers [33, 36, 42, 173, 180]. The expected trend of a significant increase in VOC levels with tumor staging was not observed [28, 40, 70, 174], however, some

VOC biomarkers showed statistical differences between stages [37, 54, 83, 86, 171, 172, 175, 177–179]. Amal et al. showed that breath test model can be used to distinguish early and late stage ovarian cancer (AUC = 0.80) [170], and similar models had also been established in hepatocellular carcinoma and lung cancer [54, 84]. Nevertheless, whether breath test for tumors at different stages affects diagnostic accuracy has been studied in breast cancer, gastric cancer, and colorectal cancer, and no high-quality data have been provided nor agreed conclusions have been reached [36, 42, 173].

Performance comparison of lung cancer and non-lung cancer

To date, the predominant application of VOCs analysis has focused on lung cancer detection, likely owing to the direct anatomical relationship between exhaled breath and the respiratory system. This biological plausibility has led to ongoing scientific debate regarding the validity of VOCs-based detection for non-pulmonary malignancies. In our study, we systematically addressed this concern by conducting a comparative efficacy analysis between lung and non-lung cancers (Supplementary Fig. 5). Our meta-analysis revealed robust diagnostic performance for both cancer types, with lung cancer demonstrating a mean AUC of 0.92 (95% CI 0.89–0.92) and non-lung cancers showing comparable accuracy (mean AUC: 0.90, 95% CI 0.86–0.90). Importantly, statistical comparison showed no significant difference in diagnostic efficacy between these groups ($p = 0.115$), supporting the potential broader application of VOCs analysis beyond pulmonary malignancies.

Discussion

The detection and analysis of VOC biomarkers from exhaled breath are recognized as a promising frontier of tumor diagnosis. The primary outcome of our meta-analysis demonstrated the accuracy of VOCs-based breath test analysis for cancer diagnosis as a mean AUC of 0.94 (95% CI 0.91–0.96), a pooled sensitivity of 89% (95% CI 87%–90%), and a pooled specificity of 87% (95% CI 84%–88%). Our findings provided evidence for the high accuracy of overall VOCs-based cancer diagnosis. Our results were accordant with the previous studies: Hanna et al. did a meta-analysis on MS-based studies on all cancer types and reported a mean (SE) AUC of 0.94 (0.01), sensitivity of 79% (95% CI 77%–81%), and specificity of 89% (95% CI 88%–90%) [21]. Proved by the accordant results of previous evidence-based studies [19, 20] and our study, the accuracy of VOCs-based breath test in cancer diagnosis is encouraging. This meta-analysis establishes standardized VOC profiling protocols, identifies

cancer-specific biomarkers, and resolves methodological disparities between MS and sensor-based detection, advancing breath test clinical translation.

MS and sensor pattern recognition represent the two predominant methodologies for breath testing, each offering distinct advantages in various contexts. MS, characterized by its clearer operational principles, holds significant potential for application in scientific research, particularly in the identification of novel VOC markers and the investigation of the biological origins of known markers. Conversely, sensors are more adept for routine monitoring and large-scale screening due to their portability, cost-effectiveness, and ease of use. However, previous studies have raised concerns regarding the accuracy of sensor-based measurements, thereby raising concerns about the evaluation of diagnostic efficacy.

The primary subgroup analysis of our study investigated the performance of VOCs-based diagnosis by MS and sensors. We found that the diagnostic accuracy is not affected by whether the test is performed on MS or not (AUC 0.91 vs 0.93, $p = 0.286$). Given the significant expense and lack of portability associated with MS, sensors appear to offer greater potential for application, provided that diagnostic efficiency is maintained. However, it is important to note that the contributions of MS to theoretical research remain unparalleled by sensor technology. Also, the homogeneous sensor does not show any disadvantage in diagnostic performance compared to the heterogeneous sensor (AUC 0.92 vs 0.95, $p = 0.245$). This appears to demonstrate that the composition of the sensor can be streamlined. Certainly, this conclusion requires validation through more extensive research.

In the evaluation of diagnostic efficacy for specific cancer types, it has been observed that GC-MS, recognized as the gold standard for the detection of VOCs, demonstrates consistent performance across various cancer types. The diagnostic efficacy of the sensor is comparable to that of GC-MS in the majority of instances. Following the application of the p -test, it was determined that HPPI-TOFMS exhibited commendable diagnostic efficacy in the context of breast cancer. Conversely, the diagnostic performance of SIFT-MS in lung cancer was found to be inadequate. It is important to note that due to the limitations regarding the quantity and scale of the studies conducted, the aforementioned conclusions necessitate further validation.

Our study highlights breath VOC analysis as a complementary diagnostic tool with distinct advantages: non-invasiveness, lower cost, and point-of-care potential compared to liquid biopsy [181, 182] and Raman spectroscopy [183]. While demonstrating comparable early-stage detection capability to imaging modalities without radiation exposure, breath testing currently trails liquid

biopsy in validated biomarkers (20–30 VOC markers vs 50+ FDA-approved targets) [182]. Key strengths include superior patient compliance, population screening scalability, and real-time analysis potential. Supplementary Table S7 outlines optimal applications: breath tests for initial screening/therapy monitoring versus liquid biopsy for advanced cancer characterization. Future integration of these modalities may combine their complementary strengths while overcoming current limitations in VOC biomarkers validation.

While demonstrating clinical promise, VOCs-based breath tests currently lack FDA/EMA approval for cancer diagnosis due to unmet validation requirements—most studies involve <1000 cases with 15–30% inter-device variability, compounded by environmental interference (30% false positives) and metabolic variations (25–40% fluctuation). Compared to FDA-approved liquid biopsy (50+ markers), only 20–30 VOC biomarkers are validated. Regulatory progress is emerging: Owlstone Medical's lung cancer detector leads FDA Breakthrough Device Program (2025 pivotal trial completion), while EMA classifies devices as IVDR Class C requiring 10,000 + case evidence. With 85–92% specificity in recent trials and <\$200/test cost, VOCs analysis may soon complement existing diagnostics, particularly for lung/colorectal cancers where regulatory submissions are projected within 3–5 years. International standardization efforts (IABR initiative) remain critical to overcome current technical and evidentiary barriers for full clinical adoption.

To enhance the clinical applicability, future studies on VOCs-based breath test for cancer diagnosis should be carefully designed and reported. These studies should prioritize reliability, reproducibility, conformity, and accuracy in their diagnostic outcomes. To achieve a refined VOCs collection process, a standardized VOCs analysis, and optimized diagnostic modeling, we proposed a workflow and reporting framework for VOCs-based breath analysis studies in cancer diagnosis. Such framework was based on the published initiatives for VOCs study standardization reported by the International Association of Breath Research and the European Respiratory Society and the observed influencing factors in the quality assessment [7, 184].

Our meta-analysis had several limitations. First, most of the included studies are preclinical studies that are single-center-based, with limited sample sizes and unstandardized workflow, which may increase heterogeneity. Secondly, many of the included studies lack an external validation with the risk of overfitting. Thirdly, a paucity of studies reported turnover time and economic expenditure.

Conclusions

The VOCs-based breath test shows high efficacy for cancer diagnosis. There is no significant difference between the efficacy of VOCs detection by sensors or MS. The diagnostic efficacy of an individual sensor merits acknowledgment as well. Thus, we provided a solid foundation for further application of sensor-based VOCs detection in breath tests for cancer diagnosis. Larger sample sizes and delicate protocols are still needed to standardize VOCs-based breath test procedures.

Methods

Protocol and registration

We reported this systemic review and meta-analysis based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement. The review protocol was previously registered with the registration number CRD42021234899 in PROSPERO 2021 on Apr. 7, 2021 [185].

Eligibility criteria

In the systemic review assessing methodology and technology, studies were included if they met the following requirements: (1) trials analyzing exhaled VOCs for cancer diagnosis, and (2) studies identifying VOCs with MS or sensors. Studies were excluded if: (1) canine olfaction or electronic nose detection was employed without VOCs identification, (2) VOCs were collected from matrices other than exhaled breath, such as breath condensate, saliva, urine, serum, feces, nipple aspirate fluid, or gastric content.

For the meta-analysis evaluating diagnostic accuracy and subgroup analyses, additional inclusion criteria required studies to report diagnostic accuracy for a specific cancer type compared to a healthy control. Necessary data include the number of participants, true positive number, false-positive number, false-negative number, and true negative number. Supplementary studies were retrieved from references of the included studies.

Information source and search strategy

A search was conducted on May. 1, 2025, in subsequent databases: MEDLINE via PubMed, Web of Science, EMBASE, ClinicalTrials.gov, Cochrane Central Register of Controlled Trials, and Cochrane Database of Systemic Reviews. Search terms included the keywords “cancer” AND (“volatile organic compound” or “VOC” OR “breath

test”) AND “diagnosis”, along with their synonyms. The detailed search strategy is provided in Supplementary Method S1.

Study selection, data items, and collection process

Included studies were screened by two independent reviewers according to the eligibility criteria. For each study, data were extracted on pre-sampling preparation, exhaled breath collection, fraction separation, analytical platform, discriminative VOCs, sensitivity, specificity, and the AUCs of ROC of diagnostic models. AUCs were determined from graphical ROC curves with Engauge Digitizer (version 11.1) [186].

Summary measures and synthesis of results

Data were collated from included studies for summary outcomes. Pooled analyses of diagnostic test accuracy yielded the primary outcomes of our study, which included sensitivity, specificity, and AUC of sROC. Subsequent subgroup analyses utilized dichotomous categories such as study design, breath fraction, and analysis platform. *Midas* software facilitated the synthesis of primary outcomes and comparison of subgroup sensitivity and specificity through Stata version MP 16.0 (StataCorp, TX) [187]. Comparative analysis of AUC in subgroup and leave-one-out analyses utilized R version 4.0.3 with the *dmetatool* package, applying the bootstrap method [188]. Significancer was attributed to two-sided p-value < 0.05. Statistical heterogeneity was assessed across studies using univariate tests for heterogeneity in sensitivity and specificity, along with I² statistic estimates [189].

Abbreviations

AUC	Area Under the Curve
CNTs	Carbon nanotubes
CT	Computed tomography
DCIS: ductal carcinoma in situ EBC	Exhaled breath condensate
E-noses	Electronic noses
ESI	Electrospray ionization
FT-ICR-MS	Fourier-Transform-Ion Cyclotron Resonance-Mass Spectrometry
GC-MS	Gas chromatography-mass spectrometry
GI	Gastrointestinal
HC	Healthy control
HPLC	High-performance gas chromatography
HPPI-TOFMS	High-pressure photon ionization time-of-flight mass spectrometry
IDT	Interdigitated transducer
IMS	Ion-mobility spectrometry
LRA	Logistic regression analysis
LOOCV	Leave-one-out cross-validation
LOQ	Limit of Quantification
MNPs	Metallic nanoparticles
MCNPs	Metal monolayer-capped nanoparticles
MOS	Metal oxide semiconductors
MPM	Malignant pleural mesothelioma
MS	Mass spectrometry

MWCNTs	Multi-walled carbon nanotubes
m/z	Mass-to-charge ratio
NTDs	Needle trap devices
NTs	Nanotubes
NWs	Nanowires
PLS-DA	Partial least squares discriminant analysis
ppbv	Parts per billion by volume
ppmv	Parts per million by volume
pptv	Parts per trillion by volume
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PTR-MS	Proton-transfer reaction mass spectrometry
QCM	Quartz crystal microbalance
QUADAS-2	Quality Assessment of Diagnostic Accuracy Studies 2
RF	Random forest
ROC	Receiver operating characteristic
SAW	Surface acoustic wave sensor
SERS	Surface enhanced Raman scattering
SESI-MS	Secondary electrospray ionization-mass spectrometry
SIFT	Selected ion flow tube
SPI-MS	Single-photon ionization-mass spectrometry
SPME	Solid phase microextraction
SPR	Surface Plasmon Resonance
sROC	Summary receiver operating characteristic
SVM	Support vector machine
SWCNTs	Single-walled carbon nanotubes
TD	Thermal desorption
TOF	Time-of-flight
VOCs	Volatile organic compounds
WDA	Weighted digital analysis

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12951-025-03526-4>.

Supplementary material 1

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Author contributions

Dr. Liu had full access to all the data in the study and took responsibility for the integrity of the data and the accuracy of the data analysis. Ziqi Jia: conceptualization, data curation, formal analysis, investigation, methodology, software, visualization, writing—original draft preparation, writing—review & editing. Yiwen Jiang: data curation, formal analysis, investigation, methodology, Software, visualization, writing—original draft preparation, writing—review & editing. Tongxuan Shang: conceptualization, data curation, formal analysis, methodology, Software, visualization, writing—original draft preparation, writing—review & editing. Chang Bao Han: conceptualization, project administration, resources, supervision, writing—review & editing. Jiaqi Liu: conceptualization, funding acquisition, project administration, resources, supervision, writing—review & editing. Xiang Wang: funding acquisition, project administration, resources, supervision, writing—review & editing. All authors reviewed and revised the manuscript prior to submission.

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Data availability

No datasets were generated or analysed during the current study.

Declarations**Competing interests**

The authors declare no competing interests.

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